

Question number	Answer	Mark
4(a)(i)	B The only correct answer is B. <i>A is incorrect because fructose is a monosaccharide</i> <i>C is incorrect because both glucose and fructose are monosaccharides</i> <i>D is incorrect because fructose is a monosaccharide</i>	(1)

Question number	Answer	Mark		
4(a)(ii)	C <table><tr><td>peptide</td><td>condensation</td></tr></table> The only correct answer is C. <i>A is incorrect because ester bonds are in lipids</i> <i>B is incorrect because ester bonds are in lipids and hydrolysis splits bonds</i> <i>D is incorrect because hydrolysis splits bonds</i>	peptide	condensation	(1)
peptide	condensation			

Question number	Answer	Mark		
4(a)(iii)	B <table><tr><td>absent</td><td>lower</td></tr></table> The only correct answer is B. <i>A is incorrect because saturated fatty acids have more hydrogens than unsaturated fatty acid of same length</i> <i>C is incorrect because there are no carbon carbon double bonds in saturated fatty acids</i> <i>D is incorrect because there are no carbon carbon double bonds in saturated fatty acids</i>	absent	lower	(1)
absent	lower			

Question number	Answer	Additional guidance	Mark
4(b)(i)	• volume of cylinder calculated (1) • ratio of surface area : volume to max 2 dps (1)	Example of calculation: $(3 \times 2 \times 2 \times 28 / \pi \times 2 \times 2 \times 28 / \frac{22}{7} \times 2 \times 2 \times 28) 336 / 352 \text{ (mm}^3\text{)}$ e.g 1.02 : 1 / 1.07 : 1 / 1 : 1 ACCEPT ratio given the wrong way round e.g. 1 : 09 NB ecf if {incorrect rounding from above / diameter used} for 1 mark	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes two of the following points: • (because the insect is small) the cells will not be far from the {sinuses / blood} (1) • (so) diffusion can supply the {oxygen / nutrients / named nutrient} (from the blood) (1) • because the insect is small it does not have {double circulation / blood vessels / closed circulatory system / complex circulatory system / high pressure} OR • because the insect has a low {metabolic rate / oxygen demand} (1) • diffusion can supply the oxygen (from blood / tracheoles) / the oxygenated and deoxygenated blood <u>does not need to be kept separate</u> (1) • because the insect is small it does not have {double circulation / blood vessels / closed circulatory system / complex circulatory system / high pressure}	IGNORE ref to surface area to volume ratio ACCEPT small diffusion distance between cells and {sinuses / blood} ACCEPT remove carbon dioxide diffusion supplies oxygen to cells from the tracheoles IGNORE has an open system because this is in the question IGNORE has an open system because this is in the question	(2)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes two of the following points: - fibrous protein (1) {triple / three stranded} helix (held with hydrogen bonds) (1) (short) repeating sequences of amino acids / high {hydroxyproline / proline / glycine} content / every third amino acid is glycine (1)	IGNORE refs to secondary / tertiary / quaternary structure throughout NB piece together IGNORE alpha	(2)